How-To Library

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Customer purchase insights with Open Table Connector

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Abstract

You can leverage Apache Iceberg tables through Open Table Connector to analyze evolving customer behaviors and recognize significant purchasing patterns across specified time frames.

Supported Versions

- Cloud Data Integration

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Use case overview

BrightMart is a leading retail organization specializing in a diverse range of consumer products, catering to millions of customers through both online and physical stores.

They focus on delivering a seamless shopping experience with frequent promotions, seasonal sales, and an extensive product catalog that continually evolves to meet customer demands.

Existing challenges

With large volumes of transactional data generated daily, including frequent changes like price updates, discounts, and product catalog modifications, BrightMart faces challenges in analyzing and understanding evolving customer purchasing behaviors.

In addition, due to audit requirements, BrightMart needs to generate summarized reports that reflect the state of transactional data as of an earlier time period. This historical view is essential for compliance, financial reviews, and validating past business decisions.

These complexities make it difficult for BrightMart to track evolving purchasing patterns over time and leverage this insight to make informed business decisions.

Solution

To address these challenges, BrightMart can implement a solution that leverages Apache Iceberg tables through Open Table Connector to perform time travel queries. This approach enables BrightMart to access historical snapshots of customer purchase data across specified time frames.

BrightMart's customer purchase data can be ingested and stored in Apache Iceberg tables. The underlying storage uses scalable object storage such as Amazon S3, while the Iceberg table metadata is managed through a catalog service such as the AWS Glue Catalog.

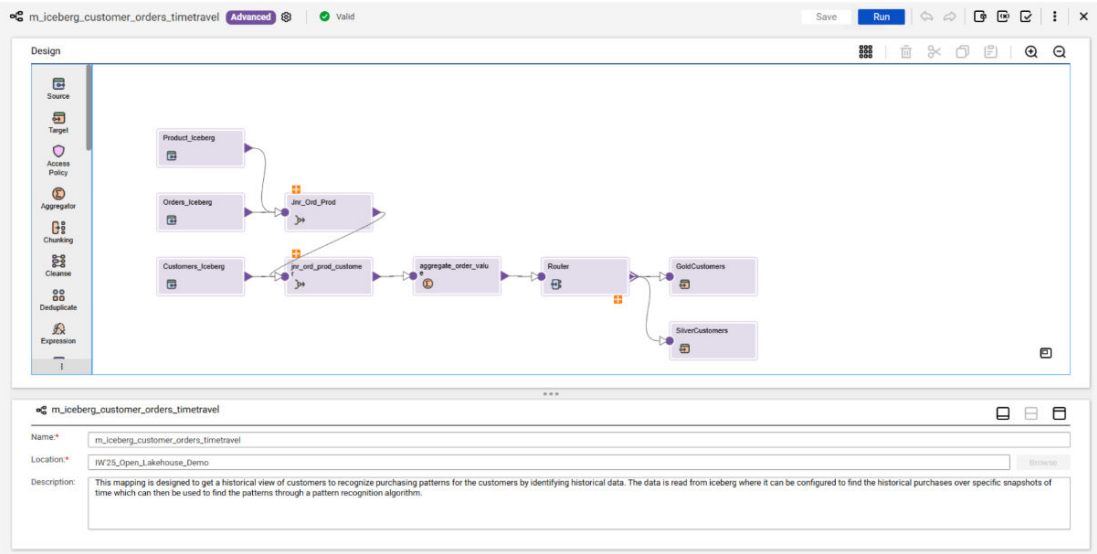
BrightMart's analysts and applications can use Open Table Connector to seamlessly query historical snapshots from Iceberg without the need to maintain multiple data copies.

After the data is processed, the aggregated and enriched purchase metrics and customer segments can be written to Iceberg tables using Open Table as the target. This design ensures data consistency, supports auditing requirements, and provides efficient access to evolving purchase patterns for reporting and strategic decision making.

Mapping configurations

To analyze historical customer purchase patterns and recognize evolving behaviours, BrightMart can configure Open Table mappings that leverage the Apache Iceberg time travel capabilities.

The following image shows a sample mapping named *m_iceberg_customer_orders_timetravel* that is designed to extract historical purchase data from Iceberg tables by querying specific snapshots in time:



This mapping uses an Open Table connection as the source in which the Apache Iceberg tables are managed by the AWS Glue Catalog with Amazon S3 storage. Multiple transformations can be used to combine data from Products, Orders, and Customers Iceberg tables and enrich the dataset.

The purchase metrics can be summarized to categorize customers based on their purchase behavior. Finally, a Target transformation can be added that uses an Open Table connection and writes the processed and aggregated data into Iceberg tables to further analyse customer purchase patterns.

Before you begin

Before you create an Open Table connection, ensure that you meet certain prerequisites.

When you use an AWS Glue Catalog and Amazon S3 storage to interact with Apache Iceberg tables, you need to have access to the following AWS services that manage the tables on AWS:

- **AWS Glue Catalog**: Manages the metadata associated with the Apache Iceberg tables.
- **Amazon S3 Storage**: Stores the Apache Iceberg tables containing actual records in columnar format, organized in partitioned directories.

- **Amazon Athena**: Uses the AWS Glue Catalog to store metadata such as table and column names for your data stored in Amazon S3. Open Table Connector uses the Amazon Athena JDBC driver to connect to the AWS Glue Catalog to access Apache Iceberg tables metadata and perform SQL queries on data stored in Amazon S3 storage.

Create minimal IAM policies

You need to create IAM policies with the minimum required permissions to interact with Apache Iceberg tables managed by AWS Glue Catalog. For more information on configuring these policies, refer to the AWS documentation.

Minimum policy for Amazon Athena

The following sample policy shows the minimal Amazon IAM policy to access Amazon Athena:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "athena:CreatePreparedStatement",
                "athena:GetPreparedStatement",
                "athena:GetWorkGroup",
                "athena:GetTableMetadata",
                "athena:StartQueryExecution",
                "athena:GetQueryResultsStream",
                "athena:ListDatabases",
                "athena:GetQueryExecution",
                "athena:GetQueryResults",
                "athena:GetDatabase",
                "athena:ListTableMetadata",
                "athena:GetDataCatalog",
                "athena:DeletePreparedStatement"
            ],
            "Resource": [
                "arn:aws:athena:*:*:workgroup/*",
                "arn:aws:athena:*:*:datacatalog/*"
            ]
        },
        {
            "Sid": "VisualEditor1",
            "Effect": "Allow",
            "Action": [
                "athena:ListDataCatalogs",
                "athena:GetQueryExecution",
                "athena:ListWorkGroups",
                "athena:GetPreparedStatement"
            ],
            "Resource": "*"
        }
    ]
}
```

Minimum policy for AWS Glue

The following sample policy shows the minimal Amazon IAM policy to access AWS Glue Catalog:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "glue:*"
            ],
            "Resource": [
                "*"
```

```
                ]
            }
        ]
    }
```

Minimum policy for AWS S3

The following sample policy shows the minimal Amazon IAM policy to read from or write data to an Amazon S3 bucket:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "s3:PutObject",
                "s3:GetObject",
                "s3:ListBucket",
                "s3:DeleteObject"
            ],
            "Resource": "*"
        },
        {
            "Effect": "Allow",
            "Action": [
                "s3:ListBucket",
                "s3:GetBucketLocation",
                "s3:ListAllMyBuckets",
                "s3:GetBucketAcl"
            ],
            "Resource": [
                "*"
            ]
        }
    ]
}
```

For information about configuring an Open Table connection, see Connect to Open Table.

Configure a Source Transformation

Configure a Source transformation that uses an Open Table connection to read the order details from the Iceberg tables containing the customer purchase data.

Open Table sources in mappings

When you configure a mapping to use an Open Table source, you can configure the source properties.

Specify the name and description of the Open Table source. Configure the source and advanced source properties for the Open Table object.

The following table describes the Open Table source properties that you can configure in a Source transformation:

Property	Description
Connection	Name of the source connection. You can select an existing connection, create a new connection, or define parameter values for the source connection property. If you want to overwrite the source connection properties at run time, select the **Allow parameter to be overridden at run time** option.
Source Type	Type of the Open Table source object. You can choose from the following source types: - **Single Object**. Select to specify a single Open Table object. - **Parameter**. Select to specify a parameter name. You can configure the source object in a mapping task associated with a mapping that uses this Source transformation.
Parameter	A parameter file where you define values that you want to update without the need to edit the task. Select an existing parameter for the source object or click **New Parameter** to define a new parameter for the source object. The **Parameter** property appears only if you select parameter as the source type. If you want to overwrite the parameter at run time, select the **Allow parameter to be overridden at run time** option. When the task runs, the Secure Agent uses the parameters from the file that you specify in the advanced session properties.
Object	Source object for the mapping.

The following table describes the Open Table query options that you can configure in a Source transformation:

Property	Description
Filter	Filter value in a read operation. Click **Configure** to add conditions to filter records and reduce the number of rows that the Secure Agent reads from the source. You can specify the following filter conditions: - **Not Parameterized**. Use a basic filter to specify the object, field, operator, and value to select specific records. - **Completely Parameterized**. Use a parameter to represent the field mapping. - **Advanced**. Use an advanced filter to define a more complex filter condition that uses the Open Table query format.
Sort	Not applicable

The following table describes the Open Table source advanced properties that you can configure in a Source transformation:

Property	Description

Iceberg Spark Properties*	The properties that you want to configure for the Iceberg tables.
	Enter the properties in the following format:
	`<parameter name>=<parameter value>`
	If you enter more than one property, enter each property in a new line.
	When you use AWS Glue Catalog with Amazon S3 and the source and target are in different regions, you must specify the Amazon S3 bucket ARN property in the following format:
	`s3.access-points.<bucket name>=<S3 bucket ARN>`
	For more information about Amazon S3 bucket ARN, see the Create the ARN for your AWS S3 bucket Knowledge Base article.
	When you use Hive metastore with Amazon S3 and the source and target are in different regions, you must specify the bucket region property in the following format:
	`BucketRegion=<Amazon-S3-source-bucket-region-name>`
	When you use REST Catlog with Amazon S3 and the source and target are in different regions, you must specify the bucket region property in the following format:
	`restcatalog.iceberg.s3.client.region = <Amazon-S3-source-bucket-region-name>`

Delta Spark Properties*	The properties that you want to configure for the Delta Lake tables.
	Enter the properties in the following format:
	`<parameter name>=<parameter value>`
	If you enter more than one property, enter each property in a new line.
	When you use AWS Glue Catalog with Amazon S3 and the source and target are in different regions, you must specify the bucket region property in the following format:
	`BucketRegion=<bucket-region-name>`

Pre-SQL	Pre-SQL queries to run before reading data to Apache Iceberg Open Table formats. You can enter multiple queries separated by a semicolon. - For mappings, ensure that the SQL queries use a valid Athena SQL syntax. - For mappings in advanced mode, ensure that the SQL queries use a valid Spark SQL syntax. You must prefix the table name with the `<OpenTableCatalog>` string and the database name. For example, `<OpenTableCatalog>.databasename.tablename` The database name and table name identifiers in the queries must be in lowercase. If the pre-SQL query fails, the mapping also fails.
Post-SQL	Post-SQL queries to run after reading data to Apache Iceberg Open Table formats. You can enter multiple queries separated by a semicolon. - For mappings, ensure that the SQL queries use a valid Athena SQL syntax. - For mappings in advanced mode, ensure that the SQL queries use a valid Spark SQL syntax. You must prefix the table name with the `<OpenTableCatalog>` string and the database name. For example, `<OpenTableCatalog>.databasename.tablename` The database name and table name identifiers in the queries must be in lowercase. If the post-SQL query fails, the mapping also fails. If the mapping fails, the post-SQL query is not executed.

Property	Description
Tracing Level	Sets the amount of detail that appears in the log file. You can choose terse, normal, verbose initialization, or verbose data. Default is normal.
*Applies only to mappings in advanced mode.	

Auditing historial data using time travel queries

You can leverage time travel queries to audit data from a specific snapshot timeframe without the need to manually maintain duplicate datasets.

This approach ensures data integrity and simplifies compliance reporting by allowing access to data exactly as it existed at a particular point or interval in the past.

When you configure a source transformation in a mapping, you can use the advanced filter option in your Open Table object to configure a time travel query. The query fetches the data based on a timestamp or snapshot ID you specify.

You can use one of the following queries to retrieve the data:

- **Query by Timestamp**: You can query an Iceberg table as it existed at a particular timestamp.
  For example, if you want to time travel to July 10, 1986 at 04:20:09, use the following time travel query:

  ```
  TimestampValue = '1986-07-10 04:20:09.0 UTC'
  ```

- **Query by Snapshot ID**: You can query an Iceberg table by specifying a snapshot ID. Each snapshot has a unique identifier.
  For example, if you want to time travel to a snapshot with ID 33444444553321, use the following time travel query:

  ```
  SnapshotVersion = 33444444553321
  ```

Apply other transformations

To enrich and analyze data from multiple sources, carry out a series of transformations that involve data joins, aggregations, and routing.

- **Joiner Transformation**: Configure a Joiner to combine data from two or more related heterogeneous sources such as Product_Iceberg, Orders_Iceberg, and Customers_Iceberg.
  For more information about creating a Joiner transformation, see Joiner transformation.
- **Aggregator Transformation**: Set up an Aggregator to calculate aggregates such as aggregate order values or relevant metrics. You can also use an Aggregator transformation to remove duplicate rows. This summarizes customer purchases to detect patterns.
  For more information about using an Aggregator transformation in a mapping, see Aggregator transformation.
- **Router Transformation**: Use a router with filter conditions that evaluates each row of incoming data. The routers or filters help to categorize customers based on purchase behavior or aggregate metrics. For example, GoldCustomers or SilverCustomers.
  For more information about how to use filter conditions in a Router transformation, see Router transformation.

Configure a Target Transformation

Configure a Target transformation that uses an Open Table connection and writes the processed historical and aggregated customer data back into an Iceberg table for further analysis or reporting.

Open Table targets in mappings

When you configure a mapping to use an Open Table target, configure the target properties.

Specify the name and description of the Open Table target. Configure the target and advanced target properties for the Open Table object in a Target transformation.

The following table describes the properties that you can configure in a Target transformation:

Property	Description
Connection	Name of the target connection. You can select an existing connection, create a new connection, or define parameter values for the target connection property. If you want to overwrite the target connection properties at run time, select the **Allow parameter to be overridden at run time** option.
Target Type	Type of the Open Table target object. You can choose from the following target types: - **Single Object**. Select to specify a single Open Table object. - **Parameter**. Select to specify a parameter name. You can configure the target object in a mapping task associated with a mapping that uses this Target transformation.
Parameter	A parameter file where you define values that you want to update without having to edit the task. Select an existing parameter for the target object or click **New Parameter** to define a new parameter for the target object. The **Parameter** property appears only if you select parameter as the target type. If you want to overwrite the target object at run time, select the **Allow parameter to be overridden at run time** option. When the task runs, the Secure Agent uses the parameters from the file that you specify in the advanced session properties.
Object	Name of the target object. You can select an existing object from the list or create a target object at run time. For information on creating the target object at run time, see the "Create a target table at runtime for AWS Glue Catalog" on page 11 topic. **Note:** You cannot create a target object at runtime if you configure Hive Metastore catalog in the connection.
Operation	Type of the target operation. Select one of the following operations: - Insert - Update* - Upsert* - Delete* **Note:** You cannot configure the Data Driven operation on an Open Table target.

Property	Description
Update Columns	Select the unique key column as the condition field for the update, upsert, or delete operations in an Open Table target.
*Applies only to mappings in advanced mode.	

The following table describes the advanced properties that you can configure in a Target transformation:

Property	Description
Truncate Target	If enabled, the Secure Agent truncates the target table before loading the data. By default, the checkbox is not selected.

Iceberg Spark Properties*	The properties that you want to configure for the Iceberg tables.
	Enter the properties in the following format:
	`<parameter name>=<parameter value>`
	If you enter more than one property, enter each property in a new line.
	When you use AWS Glue Catalog with Amazon S3 and the source and target are in different regions, you must specify the bucket region property in the following format for the update, upsert, or delete target operations:
	`BucketRegion=<Amazon-S3-target-bucket-region-name>`
	When you use Hive metastore with Amazon S3 and the source and target are in different regions, you must specify the bucket region property in the following format:
	`BucketRegion=<Amazon-S3-target-bucket-region-name>`
	When you use REST Catlog with Amazon S3 and the source and target are in different regions, you must specify the bucket region property in the following format:
	`restcatalog.iceberg.s3.client.region = <Amazon-S3-source-bucket-region-name>`

Delta Spark Properties*	The properties that you want to configure for the Delta Lake tables.
	Enter the properties in the following format:
	`<parameter name>=<parameter value>`
	If you enter more than one property, enter each property in a new line.
	When you use AWS Glue Catalog with Amazon S3 and the source and target are in different regions, you must specify the bucket region property in the following format:
	`BucketRegion=<bucket-region-name>`

UpdateMode	Loads data to the target based on the mode you specify. This property applies when you select the **Update** or **Upsert** operation. Select one of the following modes: - Update As Update. Updates records in the target table if the specified unique key column value matches with the incoming column value. - Update Else Insert. Updates records in the target table if the specified unique key column value matches with the incoming column value. If the unique key column value does not match, the mapping inserts a new row with the records.

Property	Description
Pre-SQL	Pre-SQL queries to run before writing data to Apache Iceberg Open Table formats. You can enter multiple queries separated by a semicolon. - For mappings, ensure that the SQL queries use a valid Athena SQL syntax. - For mappings in advanced mode, ensure that the SQL queries use a valid Spark SQL syntax. You must prefix the table name with the `<OpenTableCatalog>` string and the database name. For example, `<OpenTableCatalog>.databasename.tablename` The database name and table name identifiers in the queries must be in lowercase. If the pre-SQL query fails, the mapping also fails.
Post-SQL	Post-SQL queries to run after writing data to Apache Iceberg Open Table formats. You can enter multiple queries separated by a semicolon. - For mappings, ensure that the SQL queries use a valid Athena SQL syntax. - For mappings in advanced mode, ensure that the SQL queries use a valid Spark SQL syntax. You must prefix the table name with the `<OpenTableCatalog>` string and the database name. For example, `<OpenTableCatalog>.databasename.tablename` The database name and table name identifiers in the queries must be in lowercase. If the post-SQL query fails, the mapping also fails. If the mapping fails, the post-SQL query is not executed.
*Applies only to mappings in advanced mode.	

Create a target table at runtime for AWS Glue Catalog

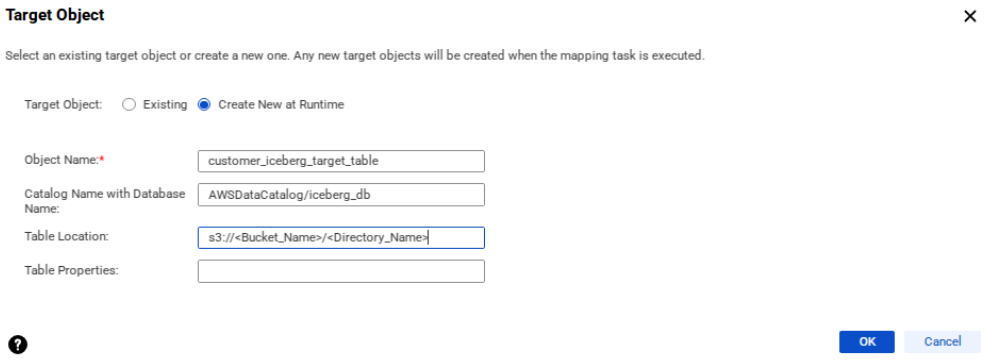
You can use an existing target or create a target to hold the results of a mapping. If you choose to create the target, the agent creates the target if it does not exist already when you run the task.

To specify the target properties, perform the following tasks:

1. Select the Target transformation in the mapping.
2. To specify the target, click the **Target** tab.
3. Select the target connection.
4. For the target type, choose **Single Object** or **Parameter**.
5. Specify the target object or parameter.
6. To specify a target object, perform the following tasks:
    a. Click **Select** and choose a target object. You can select an existing target object or create a new target object at runtime.
    b. To create a target object at runtime, select **Create New at Runtime**.
    c. In the **Object Name** field, enter the name of the target table that you want to create. The target table name must be in lowercase.
    d. In the **Catalog Name with Database Name** field, enter the AWS Glue catalog name and database name in the following format: `<catalogname>/<databasename>`. The database name must be in lowercase.
    e. In the **Table Location** field, enter the location of the target table data.
    f. Specify the table properties to optimize the table configuration settings.

g. Click **OK**.

The following image shows a sample target object:



Business benefits

Harnessing the power of its vast transactional data, BrightMart gains actionable insights that deliver significant business benefits:

- **Reduced total cost of ownership (TCO)**: By enabling direct data access at the data lake, BrightMart minimizes data movement and duplication, leading to lower infrastructure and operational costs.
- **Faster onboarding to Data Lakehouse architecture**: The integration with familiar tools and standards accelerates the adoption of a modern data lakehouse architecture, reducing time-to-value and simplifying the transition for data teams.
- **Accurate historical analysis**: The ability to perform time travel queries on Apache Iceberg tables enables BrightMart to access precise historical snapshots of customer purchase data, supporting accurate analysis of evolving buying patterns.
- **Enhanced customer insights**: By joining and enriching data from multiple Iceberg datasets, products, orders, and customers, BrightMart can identify significant purchasing trends and segment customers effectively, improving targeted marketing and customer engagement.
- **Scalable and efficient data management**: Leveraging Amazon S3 for storage and AWS Glue as the metadata catalog ensures highly scalable, cost-effective, and performant data handling suitable for large volumes generated by BrightMart's retail operations.
- **Data access with SQL**: Open Table Connector brings the familiarity and simplicity of SQL querying directly on Iceberg tables in the data lake, enabling analysts to perform complex queries, including time travel, without requiring specialized knowledge of distributed data formats.
- **Improved data consistency and auditing**: Apache Iceberg's snapshot mechanism preserves all historical data versions, which aids in auditing and compliance by allowing analysts to reconstruct data states at any point in time.
- **Operational efficiency**: Automating extraction, transformation, and loading processes through Open Table Connector reduces manual effort and accelerates data availability for reporting and analytics.
- **Support for real-time business decision making**: With timely access to aggregated and enriched historical data, BrightMart can optimize inventory management, pricing strategies, and promotional campaigns driven by data-driven insights.

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